

DS	Session	Linked List	Question Information	Level 1	Challenge 34
Question description Problem Description: For Engineering Second -year students, the instructor is organising a surprise test. He's given you a new node is added before the given node of the given Linked List. For example if the given Linked List is 5->10->15->20->25 and we add an item 30 before node 15, then the Linked List becomes 5->10->30->15->20->25. Since a Linked List is typically represented by the head of it, we have to traverse the list till node and then insert the node. Constraints: 1 < arr <100 Input Format First line contains the number of datas- N. Second line contains N integers(the given linked list). Third line contains the node P before which the node to be inserted. Fourth line contain the node X to be inserted. Output Format case 1 (node P found in Linked list) :					

Display the final Linked List.

case 2(node P not found in Linked list) :

Print Node not found!

Display the Linked list.

Logical Test Cases

Test Case 1

INPUT (STDIN)

4
115 210 116 110
116
17

EXPECTED OUTPUT

Linked List:->115->210->17->116->110

Test Case 2

INPUT (STDIN)

4
15 20 16 10
25
17

EXPECTED OUTPUT

Node not found!
Linked List:->15->20->16->10

Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

node *next;

Test Case 3

KEYWORD

*start

Test Case 4

KEYWORD

p2->next=p1;

Test Case 5

KEYWORD

void display()

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

66

LinkedList4.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node *start = NULL;
10
11 void create(int val) {
12     struct node *temp = (struct node*)malloc(sizeof(struct node));
13     temp->data = val;
14     temp->next = NULL;
15     if (start == NULL) {
16         start = temp;
17     } else {
18         struct node *p = start;
19         while (p->next != NULL) {
20             p = p->next;
21         }
22         p->next = temp;
23     }
24 }
25
```

```

26 void insertBefore(int target, int newVal) {
27     if (start == NULL) {
28         printf("Node not found!\n");
29         return;
30     }
31
32     struct node *p1 = (struct node*)malloc(sizeof(struct node));
33     p1->data = newVal;
34     p1->next = NULL;
35
36     if (start->data == target) {
37         p1->next = start;
38         start = p1;
39         return;

```

```
40 }
```

```
42 struct node *p2 = start;
43 while (p2->next != NULL && p2->next->data != target) {
44     p2 = p2->next;
45 }
46
47 if (p2->next == NULL) {
48     printf("Node not found!\n");
49     free(p1);
50 } else {
51     p1->next = p2->next;
52     p2->next = p1;
53 }
```

```
54}  
55  
56void display() {  
57    printf("Linked List:");  
58    struct node *p = start;  
59    while (p != NULL) {  
60        printf("->%d", p->data);  
61        p = p->next;  
62    }  
63    printf("\n");  
64}  
65  
66int main() {  
67    int n;
```

```
68     if (scanf("%d", &n) != 1) return 0;
69     for (int i = 0; i < n; i++) {
70         int val;
71         scanf("%d", &val);
72         create(val);
73     }
74     int target, newVal;
75     scanf("%d %d", &target, &newVal);
76     insertBefore(target, newVal);
77     display();
78     return 0;
79 }
80
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
4
```

```
115 210 116 110
```

```
116
```

```
17
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList4.exe
```

```
Linked List:->115->210->17->116->110
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
4
```

```
15 20 16 10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList4.exe
```

```
Node not found!
```

```
Linked List:->15->20->16->10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```